Static and kinetic friction

LEARNING TARGET I can distinguish adjustable static friction from kinetic friction.

SECTION A KEY VOCABULARY

Static f_s	Adjusts to prevent slip, up to a limit.
Limit	f_s,max = $\mu_s N$; f_s can be smaller.
Kinetic f_k	Opposes sliding; model size $\mu_k N$.
Normal N	Find the perpendicular force balance first.

SECTION B KEY EQUATIONS + UNITS

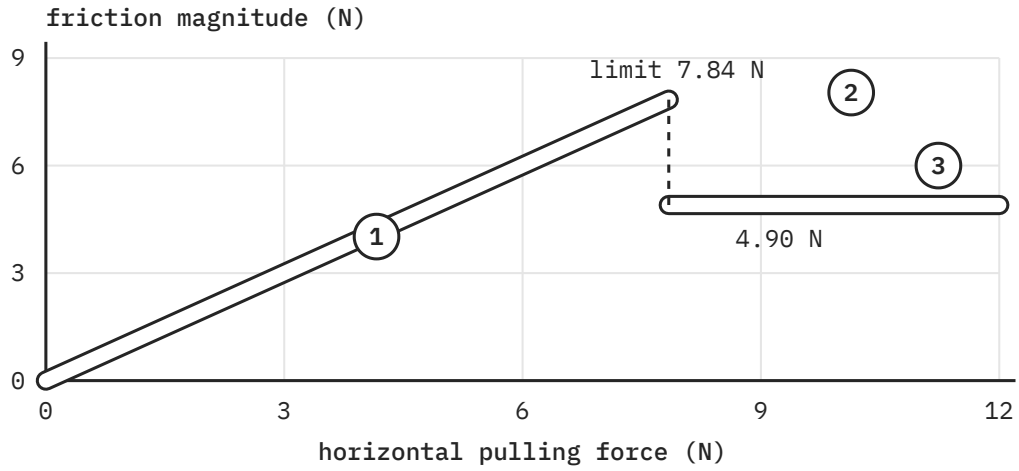
$$0 \le f_s \le \mu_s N$$

$$f_k = \mu_k N$$

μ no units
N mg here: level floor

SECTION C COLOR IT · BEFORE AND AFTER SLIP

2 kg; μ_s = .40; μ_k = .25.



☐ **1 · ORANGE** Static friction

☐ **2 · ORANGE** Static limit: **7.84 N**

☐ **3 · ORANGE** Sliding level: **4.90 N**

SECTION D WORKED EXAMPLE

For this box, compare a **6 N** pull while at rest with a **10 N** pull after sliding starts. Use g = **9.8 m/s²**; pulls are horizontal.

GIVEN	EQUATION	WORK	ANSWER
N = 2(9.8) = **19.6 N** μ_s = 0.40 μ_k = 0.25	Check $\mu_s N$. Sliding: use $\mu_k N$, then net force.	f_s,max = **7.84 N** at rest: f_s = **6 N** a = (10 – 4.9)/2	**0; 2.55 m/s²** At rest, then sliding.

SECTION E THE 2026-XJ THREAD

DAY 17 OF 22 · NEW TOOL: FRICTION STATES

Surface friction is not a force that continually brakes XJ in empty space. Gravitational forces can still change its velocity, and atmospheric drag would matter in an atmosphere.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else